MAST20030 Differential Equations

The University of Melbourne
School of Mathematics and Statistics

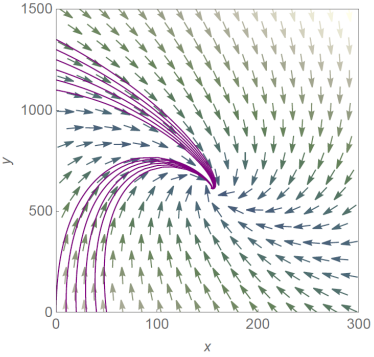
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These notes are for the use of students of the University of Melbourne enrolled in MAST20030 Differential Equations.

Original notes by Jesse F. Collis and Antoinette Tordesillas
based on material by David Ridout



Meet the MAST20030 teachers



Dr Thomas Quella
Subject Coordinator

- Lectures
- Consultions
- Canvas/LMS materials
- Tutorial Management



Dr Nick Beaton
Co-teacher

- Assessments
- Some lectures
- Ed Discussion



Rory Ward
Course Assistant

- Work behind the scenes
- LMS Quizzes?

Welcome to MAST20030 Differential Equations

- Canvas/LMS contains all info on subject
- Please read the short welcome page on Canvas/LMS and the sections on "How to approach this subject" as well as the "Subject Checklist"

Lectures

- Come to the lectures to engage with your peers and the Subject Coordinator, to listen, to ask questions and to create the positive environment that university campuses are all about!
- All lecture materials (slides, etc) will be made available online shortly after the lectures, in addition to the lectures being recorded. You can make notes during the lectures if you find that helps but there is no need to copy things verbatim.

Why study differential equations?

- Differential equations are a common approach to modelling the real world
- They provide understanding of complex phenomena and predictive capabilities

Examples in School of Maths + Stats:

- **Applied Maths:** infectious disease spread; tumour growth; bacterial interactions; lava flow; graphene; landslides; sea ice retreat; ...
- **Mathematical Physics:** string theory; quantum physics; general relativity including black holes; phase transitions; polymers; ...

In UoM Sciences:

- imaging cells; gravitational waves; ocean mixing; global warming; earthquakes; ...

Across the university:

- self-driving cars; renewable energy; nano medicine; conservation ecology; ...

Mathematical modelling of the real world in 5 steps:

1. Understand the real-world problem
2. Select the modelling approach
   - We choose differential equations as the approach. What other approaches are there?
3. Formulate the model
   - DEs, initial and/or boundary conditions, variables, parameters
4. Solve the model. Want accuracy, efficiency and simplicity $\Rightarrow$ a variety of methods
5. Interpret the solution, thereby solving the real world problem.

Case studies

We will point out potential *case studies* to learn how the methods we cover in lectures are applied to solve real-world problems.

Course outline: Six topics

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 1: Linear ODEs and IVPs

Learning outcomes:

1. How linear ODEs arise in real-world applications
2. Four methods for solving linear ODEs and IVPs
3. Connect ODEs to linear algebra
   - From Handbook: "Explain how linear algebra dictates the structure of the solution space of a linear differential equation"

Lecture 1

We will:

1. Define and classify ODEs
2. Define IVPs
3. Revise the integrating factors solution method

Poll Everywhere

https://pollev.com/thomasquella320



Linear ODEs

Definition (Classification of ODEs)

A *linear ODE* (ordinary differential equation) is a linear combination of derivatives of an unknown function $y(t)$ whose coefficients may be functions of t but not y:

$$f_n(t)y^{(n)}(t) + \ldots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

- The ODE is of order n if $f_n(t) \neq 0$.
- If $g(t) = 0$ then the ODE is *homogeneous*.
- If $g(t) \neq 0$ then the ODE is *inhomogeneous*.

Exercise 1.1: Classify the following ODEs

1. $y''(t) + \cos(t)y'(t) = t \sin t$

   Linear, second-order, inhomogeneous

2. $y'''(t) - y(t)^2 = \mathrm{e}^t \log y(t)$

   Non-linear

Initial value problems (IVPs)

Definition

1. Solutions to ODEs contain unknown constants (solutions are non-unique).
2. Constants can be fixed by specifying *initial values* at $t = t_0$ (usually $t_0 = 0$).
3. ODEs of order n require n initial values.

$$\text{Usually } y(t_0) = y_0,\ y'(t_0) = y_0',\ \ldots \text{ and } y^{(n-1)}(t_0) = y_0^{(n-1)}.$$

Exercise 1.2: Decide if the following are IVPs

1. $y''(t) = -3y(t)$ with $y(1) = 3$ and $y'(1) = 1$

   Initial value problem

2. $y''(t) = -3y(t)$ with $y(0) = 1$ and $y(2) = 0$

   Not an initial value problem

   This is a boundary value problem, which we will meet later

Integrating factor (revision)

ODEs of the form $y'(t) + f(t)y(t) = g(t)$ can be solved using an integrating factor:

$$I(t) = \mathrm{e}^{\int f(t)\,\mathrm{d}t}$$

Example: Solve

$$y'(t) - y(t) = 0$$

$$I(t) = \mathrm{e}^{\int (-1)\,\mathrm{d}t} = \mathrm{e}^{-t} \quad \Rightarrow \quad \Big(I(t)\,y(t)\Big)' = I(t) \cdot 0$$

$$\Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{-t}y) = 0 \quad \Rightarrow \quad \mathrm{e}^{-t}y = C_0 \quad \Rightarrow \quad y = C_0\mathrm{e}^{t}$$

Exercise 1.3: Find the integrating factor for the ODE

$$y'(t) - \frac{1}{t}y(t) = 1$$

In this case

$$f(t) = -\frac{1}{t}$$

and thus

$$I(t) = \mathrm{e}^{\int -\frac{1}{t}\,\mathrm{d}t} = \mathrm{e}^{-\log(t)} = \mathrm{e}^{\log(1/t)} = \frac{1}{t}$$

We now solve the problem using the IF by writing

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{t}\,y(t)\right] = \frac{1}{t}$$

and integrating both sides to give

$$\frac{1}{t}\,y(t) = \int \frac{1}{t}\,\mathrm{d}t = \log t + c \quad \Rightarrow \quad y(t) = t\log t + Ct$$